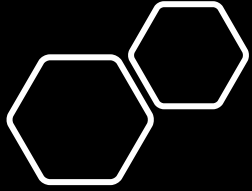


ADDRESS CLASSIFICATION

Trần Tuấn Anh



1. PROBLEM DEFINITION: Address representation

✗ Bad

ADDRESS LINE 1

4321 Sesame Street

ADDRESS LINE 2

Suite 900

2 Text Fields

Input in separate fields
Presses 'tab' for new line
Follows system format
Doesn't support all formats

✓ Good

ADDRESS

4321 Sesame Street
Suite 900

1 Text Area

All input in one field
Presses 'enter' for new line
Follows real world format
Supports international formats

Dr., Mr., Mrs., Ms.	
First Name, MI	John
Last Name	Doe
Street Address	My Street
Apt No., Suite No., Box No.	
City	My City
State or Province	MYSTATE
Zip or Postal Code	55555
Country	USA
Telephone No.	800-555-1212
Cellular Phone No.	888-555-1212
Fax No.	877-555-1212
Email	mywork@soandso.com
Email (alternate)	myhome@soandso.net

✕

Add new site

Site name

Address

find

Street number

Street

Town

City

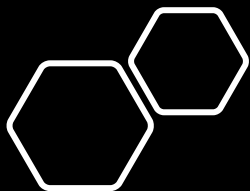
Region

Postal Code

Country

Close

Clarity of
information

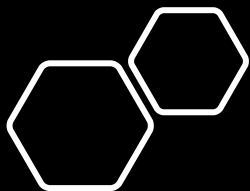


1. PROBLEM DEFINITION: Human Reading and representation



HUMAN READING &
UNDERSTANDING: confusing
rules





1. PROBLEM DEFINITION: Human & Machine Interaction

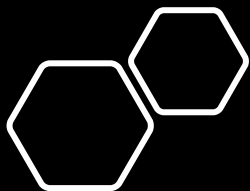
All processes
are correct?



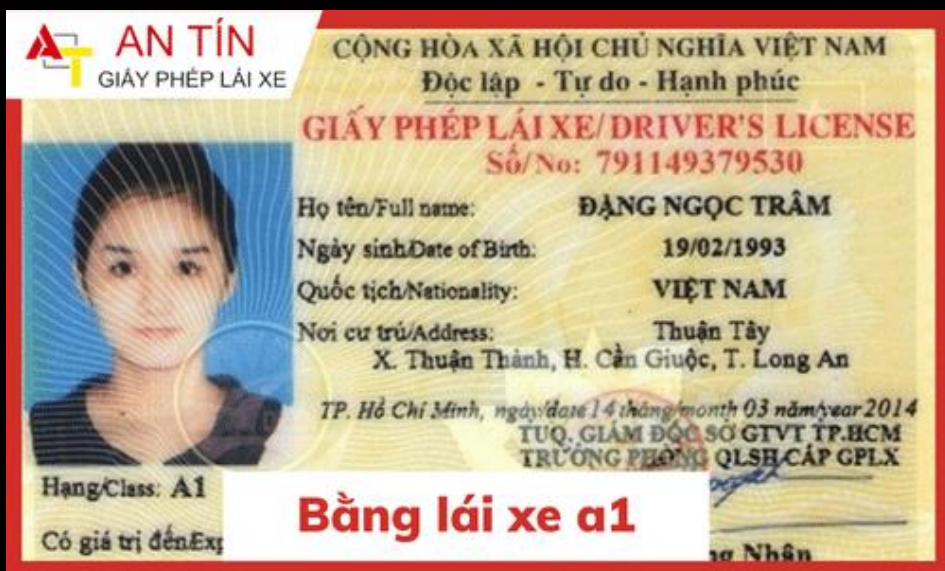
Detect address area

OCR

Classify address

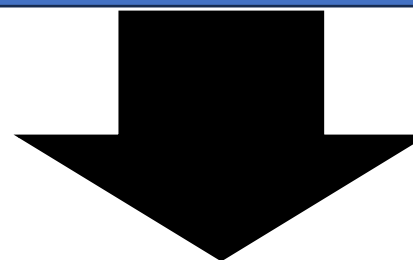


1. PROBLEM DEFINITION: Input vs Output

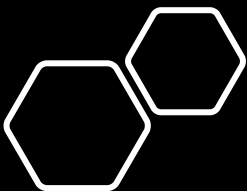


OCR:

X. Thuận Thành, H. Cần Giuộc, T. Long An
Thuận Thành, H. Cần Giuộc, T. Long An
Thuận Thành, H. Cần Giuộc, T. Long An
X Thuận Thành H. Cần Giuộc, Long An
X ThuanThanh H. Can Giuoc, Long An
.....



Address classification:
Xã: Thuận Thành
Huyện: Cần Giuộc
Tỉnh: Long An



Example

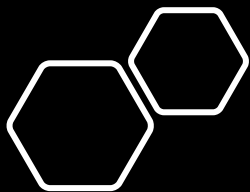
1. PROBLEM DEFINITION: Input vs Output

```
all_raw.json  X
Schema: <No Schema Selected>

1  [
2  {
3    "ground_truth": "Xã Thịnh Sơn H. Đô lương T. Nghệ An",
4    "pred_no_correct": "Xã Thịnh Sơn H., Đô lương T. Nghệ An",
5    "pred_and_correct": "Xã Thịnh Sơn, Đô Lương, T. Nghệ An",
6    "address_info": {
7      "province": "Nghệ An",
8      "district": "Đô Lương",
9      "ward": "Thịnh Sơn"
10   }
11 },
12 {
13   "ground_truth": "X.Hoi Thượng - Đồng Hỷ - Thái 2012",
14   "pred_no_correct": "X.Hoi Thượng - Đồng Hỷ- Thái 20yên",
15   "pred_and_correct": "X.Hóa Thượng",
16   "address_info": {
17     "province": "Thái Nguyên",
18     "district": "Đồng Hỷ",
19     "ward": "Hóa Thượng"
20   }
21 },
22 {
23   "ground_truth": "Tổ 73 - Hoàng Cầu - 1 Chợ Xu - Đống Đa - Hà Ni",
24   "pred_no_correct": "Tổ 73 - Hoàng Cầu -TP.1 C Xm Đống Đa - Hà Ni",
25   "pred_and_correct": "Tổ 73-Hoàng Cầu-TP.1 C Xm, Đống Đa, Hà Nội",
26   "address_info": {
27     "province": "Hà Nội",
28     "district": "Đống Đa",
29     "ward": ""
30   }
31 },
32 ]
```

Input

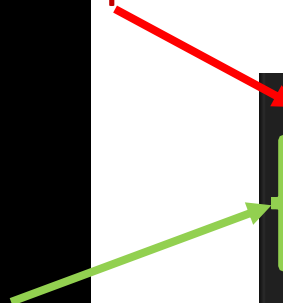
Output



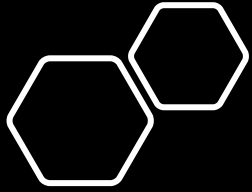
2. DEVELOPMENT & TESTING

Development provided: 1000 samples

Testing provided (private)



	all_raw	10/4/2023 3:39 AM	JSON File	579 KB
	list_district	10/4/2023 3:56 AM	Text Document	8 KB
	list_province	10/4/2023 3:56 AM	Text Document	1 KB
	list_ward	10/4/2023 3:57 AM	Text Document	97 KB
	README	10/4/2023 4:10 AM	MD File	1 KB



3. REQUIREMENTS

General:

- Submit all source code and related data.
- Readme (how to run): The Readme file should be detailed and easily understood.
- The source code will be run on a unique machine without internet.
- Any copy or cheating will get 0 for the whole course.

For the exercise:

- Maximum time for 1 request ≤ 0.2 s
- Average time for 1 request: $\leq 4/100$ s
- Duration: 1 month (private data will be released 3-4 days before the deadline; private test set is un-seen)
- Do not apply a machine learning approach, just algorithms.

